

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1/8	(a)			Conversion of 2 MHz to 2 000 000 Hz (1) Subs: $1540/2\,000\,000$ (1) $= 0.00077\text{ m} / 7.7 \times 10^{-4}\text{ m}(1)$ 7.7×10^n where n does not = -4 (2)	1	1 1		3	3	
	(b)	(i)		Iodine-123 (1) Half-life is long enough to carry out trace or short enough to decay relatively quickly (1)			2	2		
		(ii)		Cobalt-60 (1) Gamma rays will penetrate body/is the most penetrating (1) Will not need replacing often due to long(ish) half-life (1) If another gamma emitter given allow 2 nd marking point for (1)			3	3		
	(c)			R4 (1) (Lutetium is a beta and gamma emitter so) Any 2 × (1) from: paper will not affect reading / both (radiations) go through paper (1) Aluminium will reduce the reading (1) Lead will reduce the reading / there is still a reading when lead is present (1) If incorrect isotope selected, reasoning marks can still be awarded if they correspond to data in figure 6			3	3		
				Question 8 total	1	2	8	11	3	0